

Through-life Engineering Services (TES): (#1) The Value Proposition (WHY?)

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Executive Summary

Background

Why are increasing numbers of equipment and product manufacturers turning to the provision of through-life support services to enhance their market share and to access additional revenue and profit streams? Why are increasing numbers of equipment and product users buying these services, and more importantly re-buying them again and again?

The answer is that, in the right context, through-life support services add significant value to both parties by eliminating previously normalised waste in the buy, use, maintain and dispose life cycle of equipment and product 'Assets', and by placing the risk of through-life support decisions where it can best be managed.

Through-Life Engineering Services (TES) addresses the fundamental need to squeeze more value out of 'Engineered Assets' for less cost at each stage of the asset life cycle (DEVELOP usable assets, PREPARE for use, UTILISE assets and RETIRE assets from use). TES achieves this aim through the optimisation of both decisions and actions, recognising the challenges and opportunities in each life-cycle phase, and thereby reducing or eliminating previously accepted levels of waste and inefficiency.

They are often offered and delivered as a menu of service elements, enabling users to operate assets, maintaining the health of assets, upgrading assets, assuring availability of assets, optimising the value delivered by the use of assets and minimising the direct and consequential costs of using assets.

The relative value and commercial viability of these service elements varies considerably with the type of asset, its relative use, user capabilities and the life cycle stage of the asset (new to retirement). TES is therefore very varied in its scope, how it is delivered, who delivers it (internal to user organisation, or outsourced to the original equipment provider or a 3rd party service supplier) and how it is contracted. All of which may change through the life span of the assets they support.

Increasing the through-life productivity of engineered resources (e.g. physical equipment or software) is key to increasing the ability of providers, users, and owners to maximise profit from their individual stakeholder roles. This paper explores the fundamentals of the value proposition of TES, from the perspectives of the trio of stakeholder groups (asset provider, user and owner). Subsequent papers in the series will explore the types/scopes of Through-life Engineering Services and the capabilities required to deliver them.

Questions this paper addresses

- What is the basic value proposition of Through-Life Engineering?
- What is asset value?
- What is asset cost?
- Why deliver Through-Life engineering as a service?
- When is a Through-Life engineering service commercially viable?

Example value propositions

Through-life services come in many forms, comprising of many potential service elements and they may be contracted in many ways.

Below are a few examples of through-life services and their value:

Asset training:	Installation, operation, inspection or maintenance training may be offered to enable users to obtain more use, for longer and more reliably from the asset.
Technical support:	Providing expert support or continuously updated technical instructions for when users or owners have technical concerns or queries; ensuring assets are safe to use and that they are returned to use with minimum delay and cost
Monitoring:	Providing monitoring services that maintain the data required to predict the health of an asset (e.g. based on usage profile and history), monitor for symptoms of developing issues, alert users/owners when preventative action is required and provide general confidence that unexpected failures and loss of asset availability can be avoided or minimised.
Maintenance Repair & Overhaul (MRO):	Providing the physical action to inspect, maintain and potentially upgrade assets to give them additional functionality or useful failure free life.
Spares:	Providing a service level agreement to be able to supply the required spare parts or replacement assets required to keep the customers operational, with minimum non-productive down time.
Leasing:	Providing off-balance sheet options for users to obtain the functionality of an asset without the full costs of ownership
Asset planning:	Combining the knowledge of asset health, availability, operational demands and MRO availability to optimise planned maintenance for both operational productivity and cost. Planning the transition between asset life cycle phases e.g. predominantly new assets with excellent support and spares supply chains, to assets that are no longer in production (but require ongoing spares and maintenance support), to assets that are in their twilight years and no longer have any established support network.
Decommissioning & Recycling	Providing the optimal cost route to decommissioning assets, extracting the maximum residual value from their parts and minimising environmental impacts
Value analysis:	Monitoring all operational use, service support activities and costs to build a comprehensive view of additional value and cost opportunities that could be exploited.
Financial:	Providing financial services that optimise the cost and value profiles for asset users/owners e.g. pay for output, pay per use, pay per month, lease/purchase finance, shared ownership

Conclusion

Through-life engineering service (TES) has an underlying value equation:

The overall value of TES: is the incremental improvement in the net value delivered from the asset minus the cost of the TES services that release that incremental value. If the value delivered by the asset does not increase, or the costs of owning/using the asset do not decrease (and by an amount greater than the cost of the service) there is no viable TES value proposition to exploit.

$$\text{TES Value} = \Delta(\text{Asset value} - \text{asset cost}) - \text{cost of TES}$$

Where: Δ = TES created improvement

We can further break down these elements to look at the individual value levers that TES can impact:

Asset Value: The value delivered by the asset will in most cases be related to:

- the amount it is used both in terms of the quantity of use and the value delivered each time it is used.
- insurance value, where the value of the asset is in its ability to be used, even if it remains unused e.g. a medical kit or a defibrillator.
- and the asset itself will have a tangible resale value (or cost) that will depreciate (or increase).

$$\text{Asset value} = \Sigma(\text{value per use} * \text{quantity of use}) + \text{change in residual value}$$

- value per use = value from output + value from availability
- quantity of use = lifespan * availability * utilisation rate (when available)
- residual value = Transfer value of the asset to a new owner/user

Asset Cost: Assets generate several forms of cost:

- The direct costs of acquisition, day to day operation, maintenance and disposal
- The indirect costs of resources (human, physical and financial) required to retain and use them
- The costs the asset creates or influences, but which are not directly spent on the asset
 - Impacts on the environment, operational disruption, organisational reputation, lost opportunities, cost of replacement or substitution if not available for use, general inconvenience and creep in other activity costs and the costs of managing sustainability issues (material stockpiling, legacy supply chain maintenance etc), and potential obsolescence of parts (spares no longer able to replace them)
- The costs of risk and its management, including insurance and the provision of contingency resources, both human, physical and financial that may never be used but have their own costs.

$$\text{Asset cost} = \text{consumed cost (direct + indirect)} + \text{consequential cost} + \text{risk premium}$$

- Direct costs = acquisition + operation + maintenance + disposal
- Indirect costs = support resources, facilities, consumables, financial provisions
- Consequential costs = environmental + disruption + reputational + lost opportunity + substitution + inconvenience + sustainability management
- Risk premium = insurance + contingency resources (financial & physical)

Through-life engineering services have commercial potential when the proposed service provider can deliver significant incremental improvement, in one or more elements of the value equation, relative to the incumbent stakeholders in the asset life cycle (supplier, owner, user, maintainer etc).

This may involve internalising services that are currently outsourced; outsourcing current internal services or introducing services (internal or outsourced) that are not currently performed at all.

To be commercially viable over the long term, the through-life service must generate a significant incremental value (after its costs), and that value must be equitably shared between the stakeholders (service provider, customer and supply chain) through acceptable commercial arrangements.

1. The basic concept of TES is that you can optimise the total Value/Cost equation for an asset (and each of its stakeholders) by optimising decisions and actions for their total life cycle impact rather than for their short-term transactional cost.
2. It is the effective application of engineering knowledge, derived from experience (gained in the design, use and maintenance phases of the asset life cycle) that allows these optimisations to be made. It is this engineering know-how that creates the revenue opportunity from service provision, and the value proposition of consuming Through-life engineering as a service.
3. The value derived from an asset may take many forms: the value of its output (its functional purpose), value from its availability to be used (insurance, risk reduction), value from reputational enhancement (confidence building, appearance, ecological credentials etc).
4. Costs include the direct costs associated with the asset (design, acquisition, operation, monitoring, maintenance and disposal), as well as the consequential costs it creates, (cash flow risk and uncertainty, environmental, resource and facilities overheads, training, insurance, disruption of operation, loss of revenue when out of service, inconvenience, stress, uncertainty, contingency costs, reputational damage etc).
5. Through-life engineering as a services (in their many forms) have potential to exist only where the service provider (whether internal or outsourced) is clearly better placed to make (or act upon) optimised decisions throughout the assets life cycle, and can therefore generate incremental net value (better outcomes, lower cost or reduced risk for the service customer).
6. The perceived differential capability between the various stakeholders to make or act on decisions and hence incrementally increase the total value/cost equation determines the value of the service in the eyes of the service buyer.
7. The move to a commercially successful adoption of TES requires a level of mindset shift: from optimisation of immediate transactional decisions (short term cost focus) to optimisation of through-life value/cost (from the perspective of all stakeholders). In many cases the initial challenges of creating and selling a service (whether internally or externally) depend upon changing the customer mindset. They are no longer buying an asset (and all its associated costs), they are buying (to the level enabled by the services provided) its assured long-term capability, availability, utilisation potential, residual value and an isolation from the risk of volatile and uncertain costs.
8. Through-life engineering services have typically been adopted where the asset derived value/cost is high (and/or uncertain), significant opportunity for improvement exists and the

service provider (very often the original asset designer/manufacturer) is better placed to make or act upon the key through life value and cost driving decisions.

9. TES does not need to be, or should be applied uniformly to the whole of an asset, or throughout its life span. Different parts of an asset may have vastly different impacts on value generation or cost profiles and may therefore warrant entirely different services to maximise value/cost. Equally the age of an asset and/or its intended usage profile can also vastly change the potential value/cost of the services. Services should therefore be thoughtfully targeted against segmented asset sub-assemblies/parts, different usage profiles and their life cycle stage.

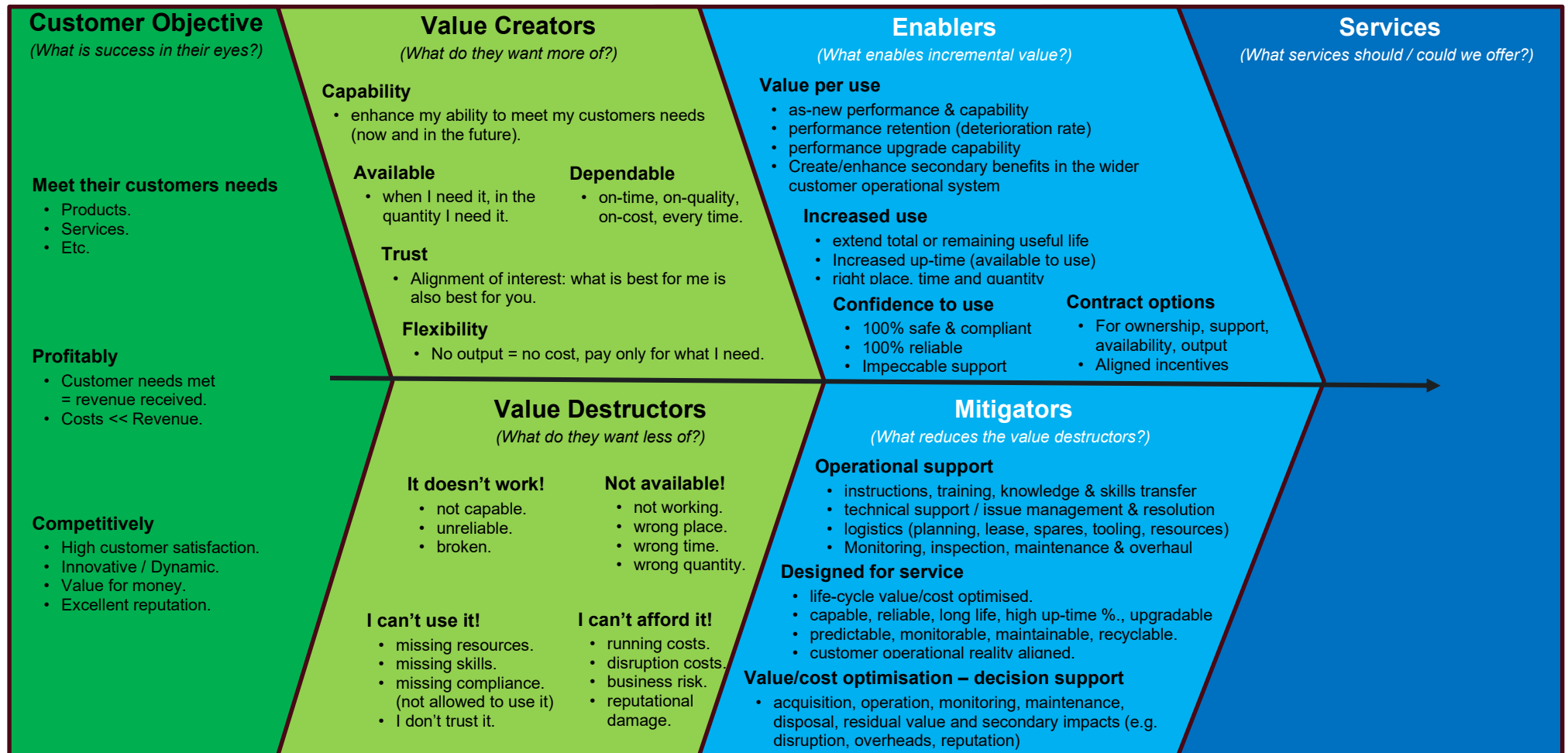
Is a through-life engineering service suitable for me?

The key questions to ask when deciding if through-life engineering services should be on your agenda are:

- Is the value derived from the asset, or the costs created by the asset high, volatile, or uncertain enough for there to be a significant value/cost to influence?
- Is there clear opportunity to optimise the value/cost profile over the assets life span through better engineering decision making (what to do, when to do it, how to do it)?
- As a result of access to better data, knowledge, capabilities, or facilities is the potential service provider better placed to:
 - make those optimisation decisions?
 - deliver the actions required by the decisions?
 - manage the associated risks?
- Can the service be delivered at a cost significantly lower than the incremental value it brings?
- Is the cost of moving to a through-life service reasonable, for both the provider and buyer?
 - Provider: creating, proving, and scaling the service.
 - Buyer: risk of change and decommissioning existing provisions.
- Is this the right time to introduce, change or remove a service due to changes in operation, user, capability or asset life cycle stage, or for convenience due to other timing factors?

Service Value Proposition

From the value discussion above we can begin to construct a service value proposition (illustrated for a business-to-business service):



The following papers in the series will expand this, discussing the potential service elements and the possible resultant service offerings.

Input Received

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The content is based upon the combined experience of many industrial and academic organisations, shared through membership and activities sponsored by the TES Council and other collaborative research and industrial peer group activities (including authoring the British Standards Institute guidance *PAS-280 – Through-life Engineering Services* and the creation of the UK national apprentice standard *Through-life engineering specialist - integrated degree - level 7*)



The Report

Brief Case Study

Through-life engineering services (TES) initially developed in high value asset, business to business contexts such as civil aerospace, where Rolls-Royce (a founding TES council member) was one of the leading innovators in TES, offering long term service agreements (contracted at a \$ per flying hour rate) for its engine products since the 1990's.

Commercial airline engines have cutting edge technology, life spans of circa 25 years of operation, up to 12 hours a day of use and as much of 80% of their through-life cost is incurred in operational support, maintenance, and overhaul. Additionally, the customer for the engine asset is highly dependent upon its operational availability. If the engine is not available to power the aircraft, the

aircraft does not fly, people or cargo are not transported, the revenue for transporting them is not received and penalties may have to be paid.

This was a perfect candidate for the equipment designer/manufacturer to provide more extensive through-life support services, leveraging their expertise in the engine design and its in-service wear out mechanisms, to optimise product capability, reliability, availability, utilisation, maintenance and lifespan, obtaining more flying hours (value out) at less cost.

Additionally, the creation of advanced service contracts where the service provider is paid a fixed number of \$'s for every hour the engine is used, rather than for every hour the engine is maintained (and therefore not working) creates a massive alignment of objectives. Both the airline operator and service provider are rewarded by higher levels of engine reliability, availability and utilisation. Time spent by the engine out of service (in maintenance) benefits neither, and severely punishes the service provider (no revenue, but high costs).

The airline also sees a smoothing of cost and reduction in cost risk, as the costs are fixed and incurred incrementally over the period of operation, rather than as very discreet, large and highly variable costs incurred every few years for major maintenance events. A significant part of the value of the service is the transfer of risk from the airline to the service provider, rather than a pure reduction in cost, or increase in hours available to fly.

3 decades on, almost every large civil aircraft engine is sold with a long-term service agreement, all the major engine manufacturers offer similar contracts in order to compete (Rolls-Royce as the service contract innovator in the sector saw a huge leap in market share), and product reliability and operational availability has improved substantially.

Since then, TES has expanded with more and more sectors seeing original equipment manufacturers (OEMs), particularly of long life, high cost and operationally critical assets move into the provision of support services along-side equipment supply.

In most cases the OEM is leveraging its deep engineering knowledge of the asset to aid in the decision making of what in service logistics, inspection, maintenance, repair and overhaul support options ensure increased utilisation of the asset at equivalent or lower cost.

Analysis

What is the basic value proposition of Through-Life Engineering?

In simple terms the value proposition of TES is to increase the net value that the asset brings over its life span, or at least the period for which a customer owns or uses it.

We optimise over a period for:

Incremental TES Value = Δ (Asset value—asset cost) - cost of TES

Optimisation is achieved through the application of engineering knowledge and capability, to make better decisions throughout the life span of the asset or to execute actions more effectively or efficiently. These decisions and actions are covered in more depth in other papers in this series, but include:

- What is the best asset design to maximise total life-cycle value/cost?
- Is the asset healthy enough to be used?
- When should the asset be monitored, inspected, maintained or retired from use?
- What monitoring, inspection, maintenance should be undertaken?
- What spares, capacity and resources should be held to enable the chosen actions to be taken and operations to continue?
- When and how should I dispose of the asset?

The basic value proposition of TES is therefore to achieve one or more of the following:

- Increase the value produced (per unit of cost).
- Decrease the cost (per unit of value).
- Insulate the customer of TES from the risks of value/cost variance.

What is asset value?

In the simplest terms:

$$\text{Value} = \text{value per use} * \text{quantity of use} + \text{change in residual value}$$

Where:

Value per use = value from output + value from availability

Quantity of use = lifespan * availability * utilisation (when available)

Residual value = Transfer value of the asset to a new owner/user

Value per use: The value offered by an asset is highly dependent upon its intended functionality and the context of its use. In simple terms there are functional values (the value produced by the asset being used) and non-functional value (the value produced by the asset even when not in active use).

For example, does a Rolex or a Timex deliver more value? Both provide the ability to tell the time, but one also provides the additional value of a heightened sense of self-worth, or projection of the image of wealth, even when no-one needs to know the time. Is the primary value of an aircraft carrier measured by the number of miles it sails and planes it launches, or by its effectiveness as a deterrent to 'bad actors', simply by being available to use? Is the power supply for a life support machine more valuable than the power supply for a television?

The important thing is to understand value from the users and/or owners perspective. Why does the user/owner actually want the asset? What does it bring them, how critical is it, and what are the attributes of the asset that are most essential to enabling it?

Quantity of use: The quantity of use has three primary factors to consider:

- Life span – is the asset expected to remain usable for 1 year, 10 years, 30 years?
 - Note that 'usable' will be assessed by the customer in comparison to alternative means of achieving the same outcomes. An asset may therefore reach the end of its useful life when it becomes uncompetitive with modern alternatives, rather than when it becomes unable to function.
- Availability – is the asset in the right place at the right time, and in the right state of health to enable it to be used?
- Utilisation – even if the asset is available, will the customer choose to use it? Are there other inhibitors to its use, e.g. lack of operator training, confidence in its state of health etc?

Residual value: at the point where the assets ownership or usage is transferred to another party, there will be a transfer value (or disposal cost), representing the residual asset value (or liability) at that point in time.

What is asset cost?

The costs we should consider are the costs 'caused' by the asset, not just those 'consumed' by it.

Cost = consumed cost (direct + indirect) + consequential cost + risk premium

Direct costs: These are the costs directly associated with each individual asset:

- Acquisition - e.g. Purchase or leasing.
- Operation – e.g. consumables, fuel/power etc.
- Maintenance – e.g. inspection, repair, refurbishment etc.
- Disposal – e.g. cost (or possibly revenue) from disposal.

Indirect costs: These are the costs associated with owning or using a collection of assets:

- Resources – e.g. trainers, operators, inspectors, maintainers, tooling.
- Facilities – e.g. storage space.
- Consumables stock – e.g. spares holding.
- Financial – e.g. Insurance.

Consequential costs: These are the costs that are caused by, but not spent on the asset:

- Environmental – the emissions impacts caused by the asset.
- Disruption – the costs of managing unreliability (or unexpected lack of availability) of the asset.
 - Including the disruption from loss of availability if the asset is unsustainable/unusable
- Reputational – the cost that the asset may cause to the owners/users image and reputation.
- Opportunity – the costs of lost revenue if the asset is unavailable to be used.
- Substitution – the cost of alternative solutions if the asset is unavailable to be used.
- Convenience – the cost of operational complexity, discomfort or stress caused by the asset.

Risk premium: the cost of contingencies to insure against unreliability or lack of availability of the asset.

- Insurance premiums

- Financial account contingencies
- Excess capacity
- Spare inventory

Why deliver Through-Life engineering as a service?

The key issue with through-life decision making is the ability to manage the uncertainty and variability of the operational context.

Every individual asset has a different life history, from its design standard, material quality, manufacturing tolerance achieved, operational use (and abuse), age, accumulated wear and tear, maintenance history and intended future use.

Historically in service support decisions are based upon safe operation criteria, which means setting rules and standards that assure safety for the worst reasonably expected usage scenario of the worst reasonably expected quality of asset.

However, in the real world, very few users operate consistently in the 'worst' way. Most treat the asset far more benignly, whilst some operators will use the asset completely out of context of its design purpose (haven't we all see many objects used as a hammer when required?).

Additionally, there are thousands of scenarios of use that the asset designers would never have predicted or analysed, which leads to a near certainty that at some point, some assets will fail in use, (given enough assets in use and variations of use). The consequential costs of incurred failures can dwarf the cost of the asset and its support (e.g. failure of a life support machine, or aircraft engine).

We should also consider commercial motivations. Is the vendor selling you spare parts, maintenance hours or repair services interested in telling you when they are not needed, or are they financially motivated to maximise their sales at the owner or users expense? Even when the service supplier is part of the same company, they may be part of a separate cost unit and therefore influenced by inter department trading rules and accounting practices that drive departmental rather than company optimised decision making.

For this collection of reasons, and in the typical commercial supplier/buyer context, there is a high degree of through-life decision making inefficiency.

- the asset manufacturer selling an asset – motivated by the initial sale, not its through-life cost.
- the owner making the use, maintain, dispose decisions – with imperfect information.
- Vendors (or internal departments) supplying inspect, repair, replace services – commercially motivated to sell as much work as possible.

This leads to the wrong actions being taken (or not taken) at the wrong times with every action creating a cost (which will be incurred) and an intended benefit (that may not).

- Too early – cost incurred, but cost-free operational time is lost.
- Too late – failures occur with potentially massive consequential costs.
- Too little – insufficient further use is achieved for the cost of the action – poor value for money.

- Too much – cost is far higher than required to achieve the same value from future use.

The result is an estimated >30% level of waste in the cost of in-service support activities for many major industrial assets with high existing support costs or criticality of use/availability.

So why deliver TES as a contracted service (even if provided internally within an organisation)?

- **Capability:** Some parties are better placed to make the key support decisions than others, or to enact those decisions more efficiently and effectively, once they are made.
 - Access to better engineering understanding of the asset (its design, use, degradation mechanisms, inspectability, repairability, remaining useful life etc) are the key ingredients to an optimised decision-making process.
 - Economy of scale, pooled resources and access to specialist equipment and skills enable the efficient and effective enactment of service decisions, typically as a service provision to a collection of owner/operator customers.
 - The decision support element is the bigger prize, as the choice of what not to do, and when not to do it can drive 100% cost reduction without negative value impact. Enactment of a decision only offers the opportunity for marginal % gains in either the effectiveness of the outcome, or the cost of the action.
- **Alignment of incentives:** in the traditional transactional model for in-service support of assets, the more support work that needs to be done, the greater the revenue and return for the service provider. In more advanced through-life services, the revenue to the service provider is mostly driven by the value delivered by the asset to the user, whilst the costs of service activity fall to the service provider. Hence both parties are aligned to wanting greater asset capability, availability (minimised down-time) and utilisation.
- **Risk transfer:** in advanced through-life engineering services, the financial risk of failing to achieve adequate asset capability or availability fall to the service provider, rather than the user. The user pays for the value obtained from the asset, rather than for ownership and/or maintenance of the asset, and their associated risks. A key value of the through-life engineering service is the placement of risk with the party most capable of predicting and managing it most cost effectively.

When is a Through-Life engineering service commercially viable?

To be commercially viable, the through-life engineering service must offer net benefits to all the service stakeholders. Their benefit-cost must be a positive and equitable outcome, compared to their other opportunities for time, effort, and resource investment.

There are 3 primary factors to consider:

1. How criticality is the asset to the users operation/activities? – are they fundamentally dependent upon it, or do they have alternative options if it is not available?
2. Are the costs of maintaining availability of the asset high (including any risk premium)? – are they significant within the users' financial model, or inconsequential?

3. Is the service provider clearly better placed to deliver better outcomes at an equitable cost? –
Increased value recovery from the asset? Decreased cost per unit of value recovery?
Decreased risk to the value/cost ratio?

At this point we must also consider who the customer is for the service. In the case of many high capital cost assets the owner may be separate to the user (who may only have access to the asset through a lease or hire purchase agreement). In that scenario, is it the user buying the service (to maximise their asset output value/cost), or the owner (seeking to maximise the assets lease revenue generating potential vs its depreciation over the period of ownership)?

In most cases there is strong alignment between the owner and users needs from a service, but with potentially different value perceptions attached to individual service elements.

The key to commercial viability is to understand the network of stakeholders, (owner, user, service supplier, service supply chain, financiers etc) and ensure that each has, and recognises a positive outcome from the move into through-life services.

Conclusion

The through-life engineering service (TES) value proposition has an underlying value equation:

$$\text{TES Value} = \Delta(\text{Asset value} - \text{asset cost}) - \text{cost of TES}$$

Where:

Δ = TES derived improvement

$$\text{Asset value} = \Sigma(\text{value per use} * \text{quantity of use}) + \text{residual value}$$

- value per use = value from output + value from availability
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- residual value = Transfer value of the asset to a new owner/user

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Through-life engineering services have potential when the proposed service provider can deliver significant incremental improvement, in one or many elements of the value equation, relative to the incumbent players in the asset life cycle (supplier, owner, user, maintainer etc).

To be commercially viable over the long term, the through-life service must generate a significant incremental value (after its costs), and that value must be equitably shared between the key stakeholders (service provider, customer and supply chain) through acceptable commercial arrangements.

